THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Another common unit of speed is: 1

(a) metre **(b)** kilometre
(c) hour **(d)** kilometre per hour

2. Earthworms breathe through their: 1

(a) moist skin **(b)** spiracles **(c)** gills **(d)** lungs

3. Sweat glands become more active in adolescence. This makes it important to: 1

(a) avoid school **(b)** avoid food
(c) avoid water **(d)** bathe and keep clean

4. In winter we prefer to wear: 1

(a) woollen clothes **(b)** net clothes **(c)** thin cotton **(d)** plastic

5. A graph showing distance covered at different times is a: 1

(a) pie chart **(b)** bar chart
(c) histogram **(d)** distance-time graph

6. Passing of digested food into the blood is called: 1

(a) excretion **(b)** absorption **(c)** egestion **(d)** ingestion

7. The chemical messengers which control changes during puberty are: 1

(a) vitamins **(b)** enzymes **(c)** minerals **(d)** hormones

8. Transfer of heat by actual movement of a fluid is: 1

(a) insulation **(b)** convection **(c)** radiation **(d)** conduction

9. One complete to-and-fro motion of a pendulum bob is called an: 1

(a) revolution **(b)** vibration of sound only
(c) rotation **(d)** oscillation

10. Bile juice is stored in the: 1

(a) stomach **(b)** liver only **(c)** gall bladder **(d)** pancreas

11. Growth of facial hair in boys is a: 1

(a) habit **(b)** secondary sexual character
(c) disease **(d)** primary sexual character

12. Water in a beaker heated from below gets heated by: 1

(a) radiation only **(b)** conduction only
(c) evaporation **(d)** convection currents

13. The formula for speed is: 1

(a) time / distance **(b)** distance / time **(c)** distance x time **(d)** distance + time

14. The organs of respiration in humans are: 1

(a) kidneys **(b)** liver **(c)** lungs **(d)** stomach

15. Pimples on the face in adolescents are caused by: 1

(a) oil glands becoming more active **(b)** eating fruits
(c) lack of sleep only **(d)** drinking water

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Differentiate between uniform and non-uniform motion. 2

17. What does haemoglobin do? 2

18. Name two secondary sexual characters in boys. 2

19. What is conduction? Give an example. 2

20. Define time period of a pendulum. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A bus covers 180 km in 4 h. Find its speed in km/h and m/s. 3

22. Explain how food is digested in the mouth and stomach. 3

23. List three habits to be developed during adolescence for good health. 3

24. Explain why a metal feels colder than wood at the same temperature. 3

OR

A train takes 5 h to cover 400 km. Find its average speed and the distance in 3 h at that speed. 3

25. What is pulse? How is it counted? 3

OR

Explain why proper sleep is essential for adolescents. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the endocrine glands, their secretions and functions with examples. 5

OR

Describe the experiment to show convection in water using potassium permanganate crystals. 5

27. A car travels at 60 km/h for 2 h, then at 40 km/h for 3 h. Find the total distance, total time and average speed. 5

OR

Explain the process of digestion of carbohydrates, proteins and fats with enzymes. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND